

Kidney Exchange Pools

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Monthly batching. At the end of each month, you have a directed graph. Maximize total matches.

Let S be the set of patient-donor pairs.

Let $C \subset S \times S$ be the set of compatible pairs. That is $(i, j) \in C$ means that the donor from pair i can give to the patient from pair j . **Represent with compatibility graph?**

We want to come up with a transplant plan $T \subset C$. It will be convenient to let this plan be represented by a set of variables $x[i][j] \in \{0, 1\}$, where $x[i][j] = 1$ means that $(i, j) \in T$ (that is, i donates to j) and $x[i][j] = 0$ means that $(i, j) \notin T$.

$$\max \sum_{i,j} x[i][j] \text{ Maximize total transplants} \quad (1)$$

$$x[i][j] \leq C[i][j] \forall i, j \in S \quad \text{Transplants can occur only between compatible patients and donors} \quad (2)$$

$$\sum_j x[i][j] \leq 1 \forall i \in S \quad \text{Each donor can give to at most one recipient} \quad (3)$$

$$\sum_j x[j][i] \leq 1 \forall i \in S \quad \text{Each patient can receive from at most one donor} \quad (4)$$

$$\sum_j x[i][j] \leq \sum_j x[j][i] \forall i \in S \quad \text{Each donor gives only if their patient receives} \quad (5)$$

$$(6)$$

This is an integer programming problem with linear constraints and objectives. **Natural question: if we relax integrality constraints**

Construct bipartite graph.

Given the directed graph G , construct the following (undirected) bipartite graph (D, P) , where D represents the set of donors and P represents the set of patients.

We have the following neat observation: if M is a perfect matching in the bipartite graph (D, P) (meaning that every vertex in M is matched), then M corresponds to a set of disjoint cycles in G . Let's do an example.

Solve maximum weight bipartite matching (i.e. with Hungarian algorithm – can go into that).

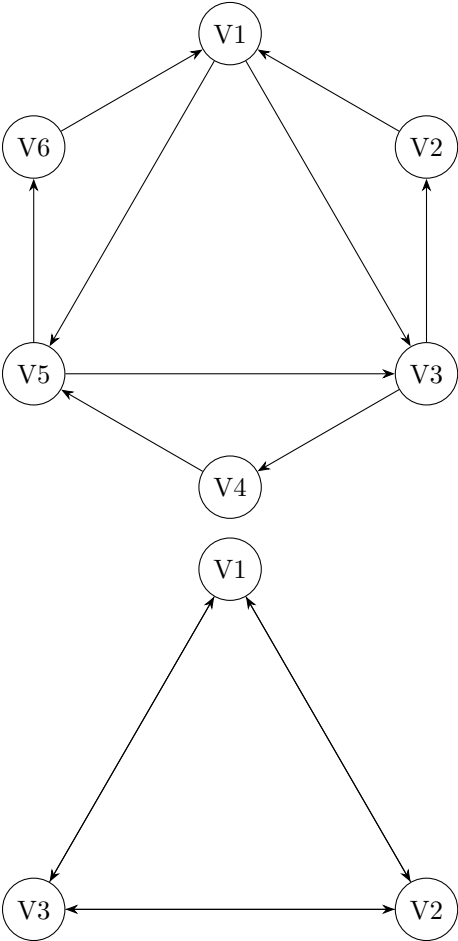
In practice, it's illegal to write a contract involving the donation of a kidney. Thus, if patient A receives a kidney before donor A gives one, donor A could in theory back out of any planned donation with no ramifications. For this reason, cycles are generally completed using simultaneous surgeries. A two-way exchange requires four surgical teams and operating rooms. A three-way exchange requires six surgical teams. For this reason, most exchanges limit the length of cycle that can be used.

Let's start with the case of 2-way cycles. We can modify the program above to add the following constraint:

$$x[i][j] = x[j][i] \quad (7)$$

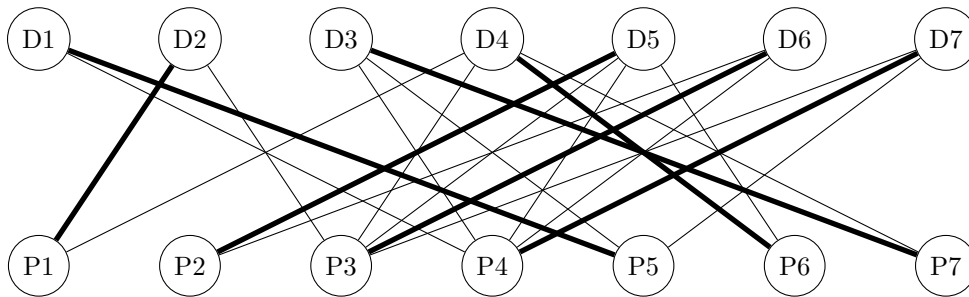
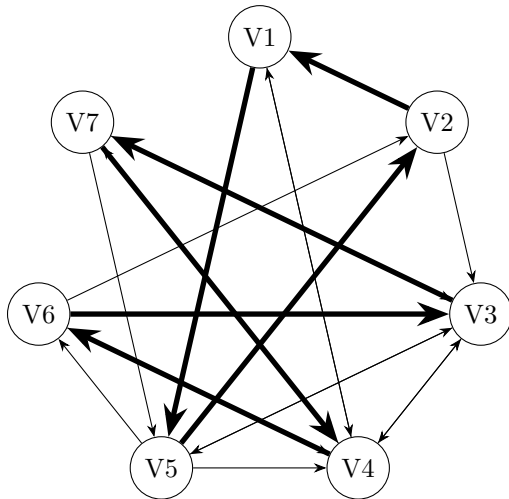
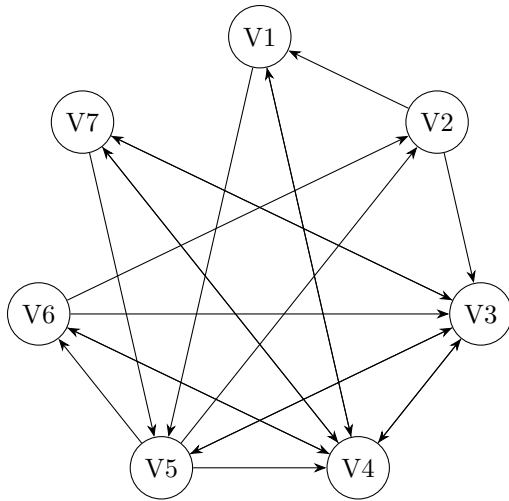
This enforces that we can only do 2-cycles by requiring that if i donates to j , then j donates to i .

Now, the relaxation into a linear program is definitely not integral. Consider the following example, with three pairs. Any donor can donate to both of the other patients. Because there are only three pairs and we are restricted to two-way exchanges, the best we can do is match two of the three pairs.



Constraint	x21	x31	x12	x32	x13	x23		
Donate 1			1		1		$\leq$	1
Donate 2	1					1	$\leq$	1
Donate 3		1		1			$\leq$	1
Receive 1	1	1					$\leq$	1
Receive 2			1	1			$\leq$	1
Receive 3					1	1	$\leq$	1
IR 1	-1	-1	1		1		$\leq$	0
IR 2	1		-1	-1		1	$\leq$	0
IR 3		1		1	-1	-1	$\leq$	0
Pairwise 1 2	-1		1				$=$	0
Pairwise 1 3		-1			1		$=$	0
Pairwise 2 3				-1		1	$=$	0

Hello World



Things to consider in optimization class:

- In general undirected graph, say that a set of agents is feasible if there is a matching that matches them all. The family of feasible sets of agents forms a matroid. Therefore (i) every Pareto efficient matching has equal size, (ii) We can use a greedy algorithm to find a max weight matching (still need to go into some details).
- Could define weight of an edge to be sum of weights of its two vertices, and then use a greedy algorithm to select edges. This will lead to a maximal matching (thus within $1/2$ of maximum cardinality matching), and also should lead to within $1/2$ of maximum weight matching.
- Matroids, greedy algorithms, linear programming, LP relaxations, hierarchy, bi-hierarchy, polynomial time, NP hardness and reductions,